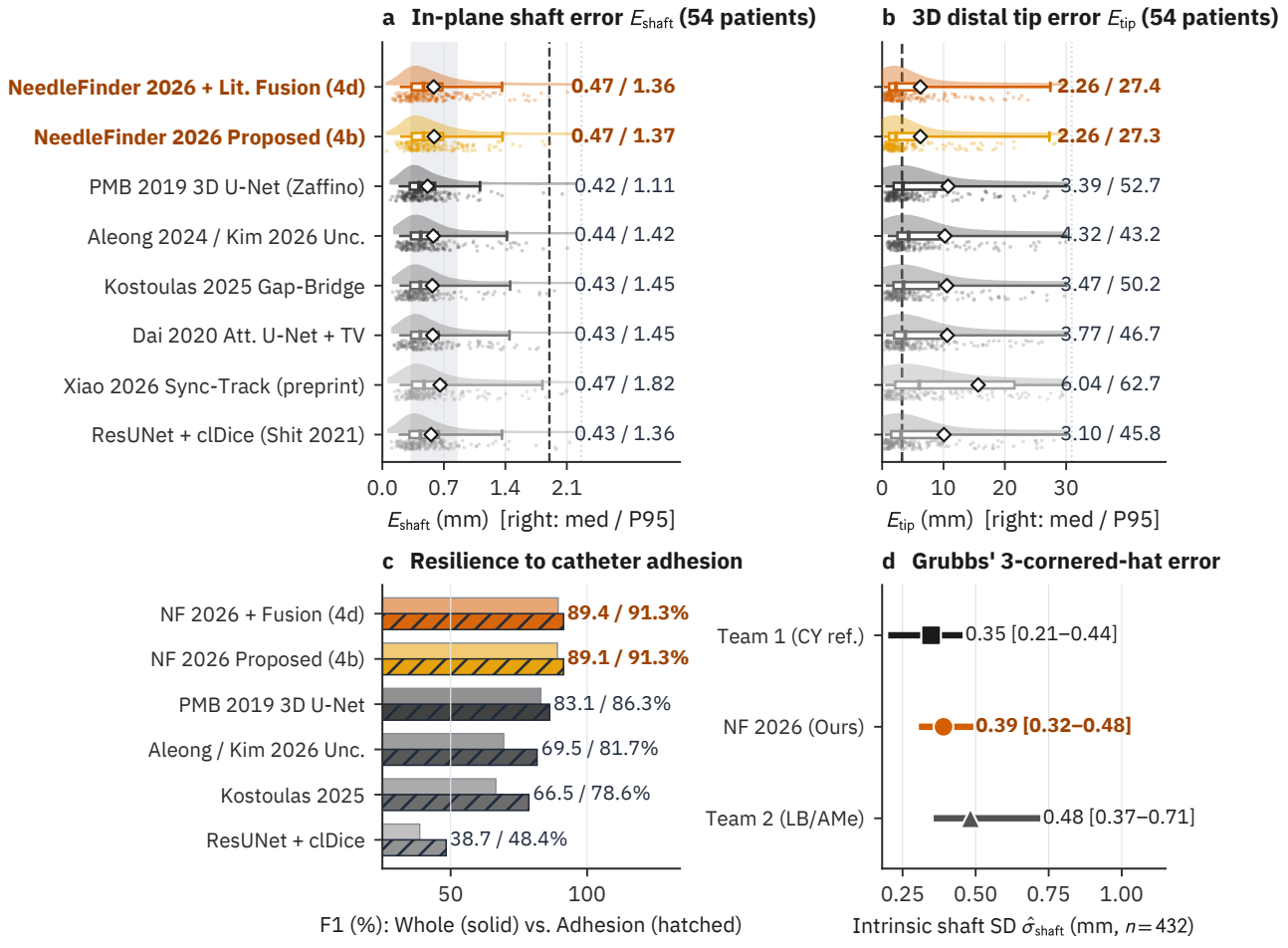

a In-plane shaft error E_{shaft} (54 patients)
NeedleFinder 2026 + Lit. Fusion (4d)
0.47 / 1.36
NeedleFinder 2026 Proposed (4b)
0.47 / 1.37
PMB 2019 3D U-Net (Zaffino)
0.42 / 1.11
Aleong 2024 / Kim 2026 Unc.
0.44 / 1.42
Kostoulas 2025 Gap-Bridge
0.43 / 1.45
Dai 2020 Att. U-Net + TV
0.43 / 1.45
Xiao 2026 Sync-Track (preprint)
0.47 / 1.82
ResUNet + clDice (Shit 2021)
0.43 / 1.36
0.0 0.7 1.4 2.1
E_{shaft} (mm) [right: med / P95]
b 3D distal tip error E_{tip} (54 patients)
2.26 / 27.4
2.26 / 27.3
3.39 / 52.7
4.32 / 43.2
3.47 / 50.2
3.77 / 46.7
6.04 / 62.7
3.10 / 45.8
0 10 20 30
E_{tip} (mm) [right: med / P95]
c Resilience to catheter adhesion
NF 2026 + Fusion (4d)
89.4 / 91.3%
NF 2026 Proposed (4b)
89.1 / 91.3%
PMB 2019 3D U-Net
83.1 / 86.3%
Aleong / Kim 2026 Unc.
69.5 / 81.7%
Kostoulas 2025
66.5 / 78.6%
ResUNet + clDice
38.7 / 48.4%
50 100
F1 (%): Whole (solid) vs. Adhesion (hatched)
d Grubbs' 3-cornered-hat error
Team 1 (CY ref.)
0.35 [0.21–0.44]
NF 2026 (Ours)
0.39 [0.32–0.48]
Team 2 (LB/AMe)
0.48 [0.37–0.71]
0.25 0.50 0.75 1.00
Intrinsic shaft SD $\hat{\sigma}_{shaft}$ (mm, $n=432$)